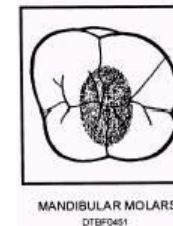
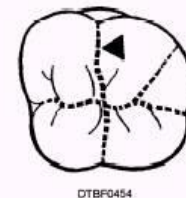
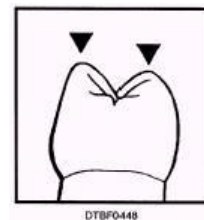
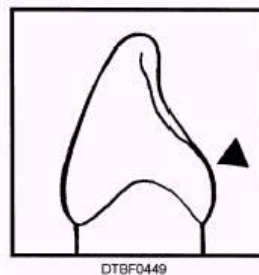
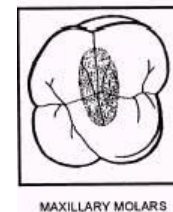
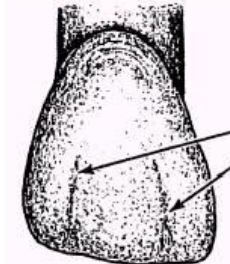
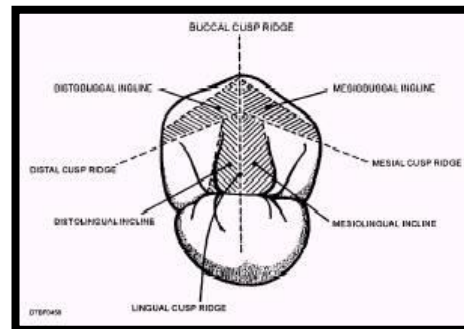
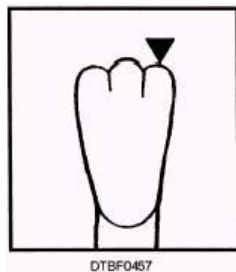
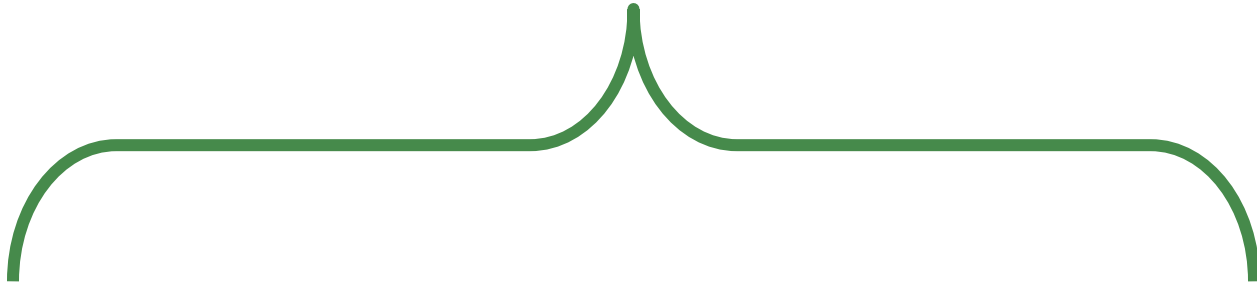


Surface anatomy

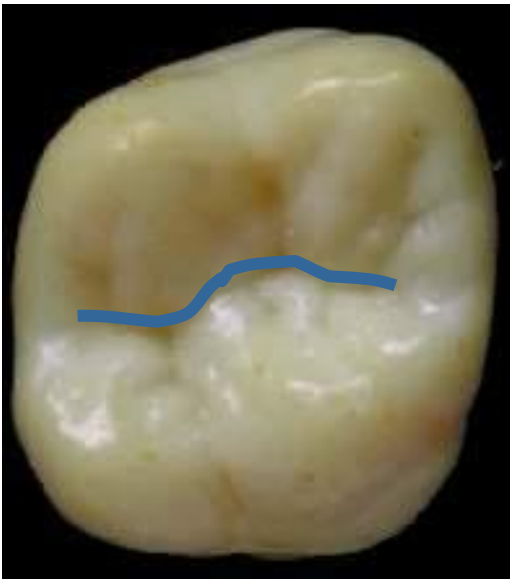
- **Surface anatomy:** The tooth surface is not a flat surface. There are elevations and depressions. Each elevation and depression is described based on its location, shape and function.



B- Crown Depressions



Linear depression



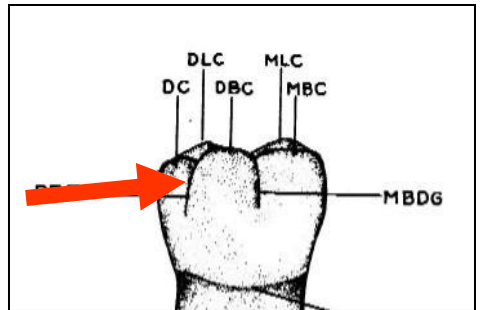
Circumscribed depressions



Linear depression

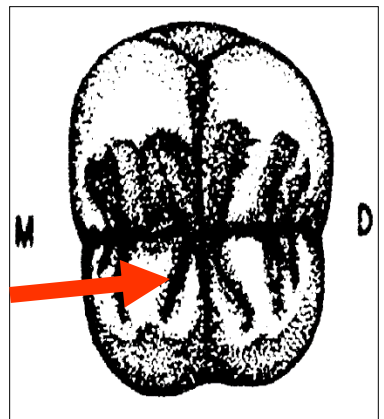
Developmental grooves

Denoting union of lobes



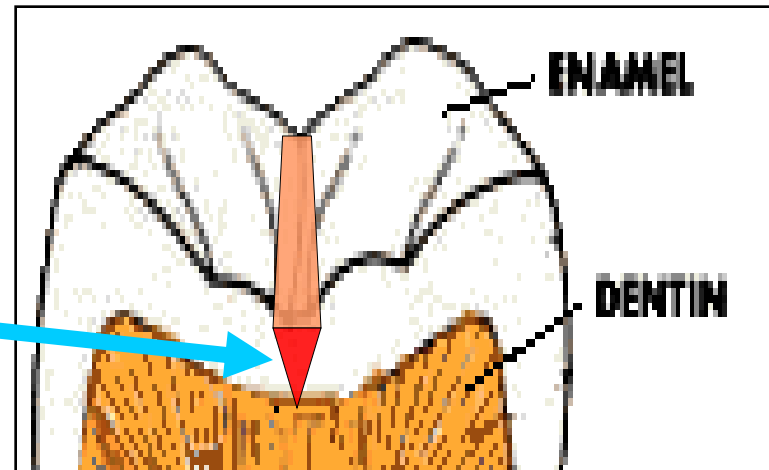
Supplemental grooves

Do not denote union of lobes



Fissures

Incomplete enamel union



1- Linear depression

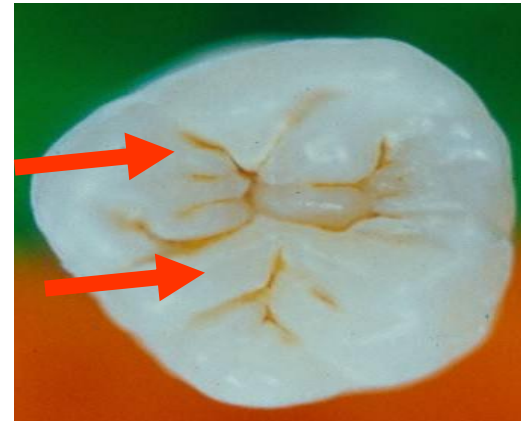
Developmental grooves

Denoting union of lobes



Supplemental grooves

Do not denote union of lobes



Fissures

Incomplete enamel union



Note:

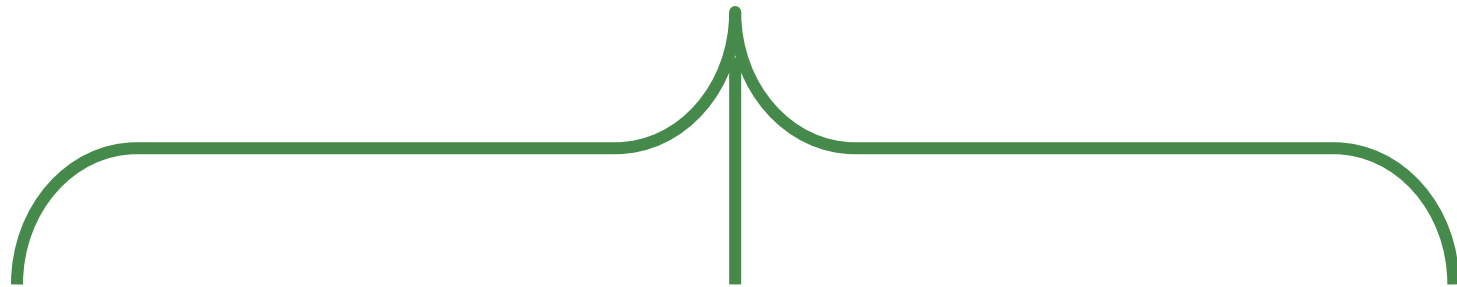
SULCUS:

It is a broad *depression or valley* on the occlusal surfaces of posterior teeth.

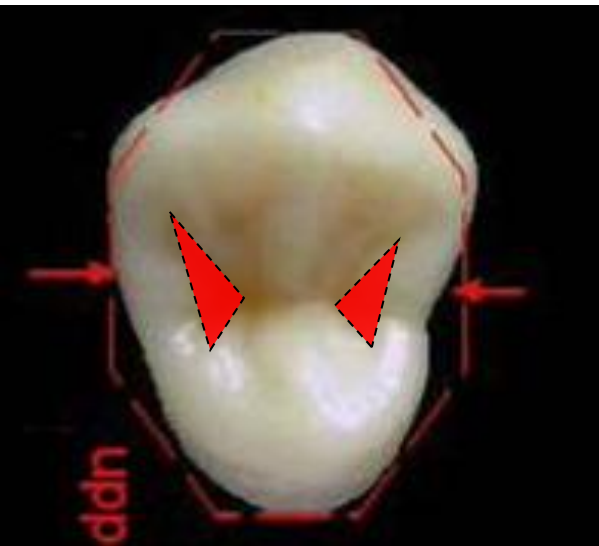
Its *inclines* meet in a developmental groove and extend to the cusp tips.



Fossa



Mesial & distal fossae



Lingual fossa



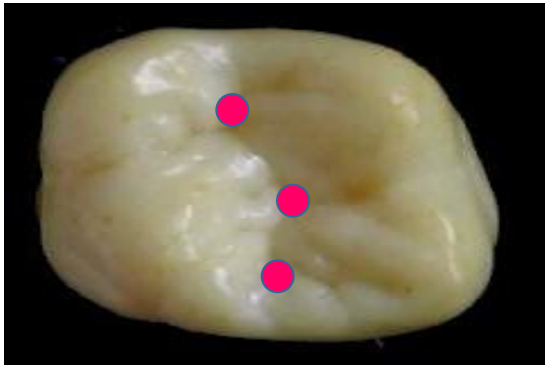
Central fossa



Pit

True pit

- Present in the bottom of central, mesial and distal triangular fossae.
- At the end of buccal groove.



Faulty pit

Usually develops as a result of incomplete formation of enamel.



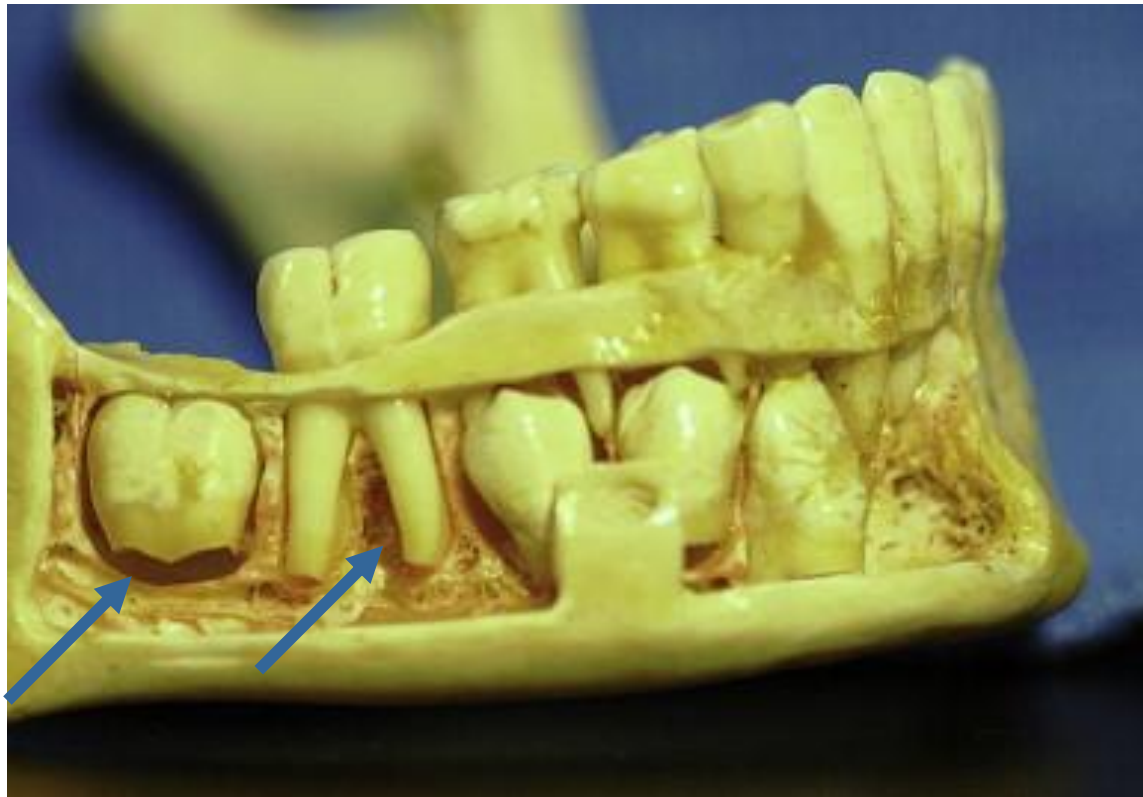




Other terms

❑ **Socket:** Bony space in alveolar bone containing roots of erupted tooth.

❑ **Crypt:** Bony space in alveolar bone containing developing unerupted tooth.

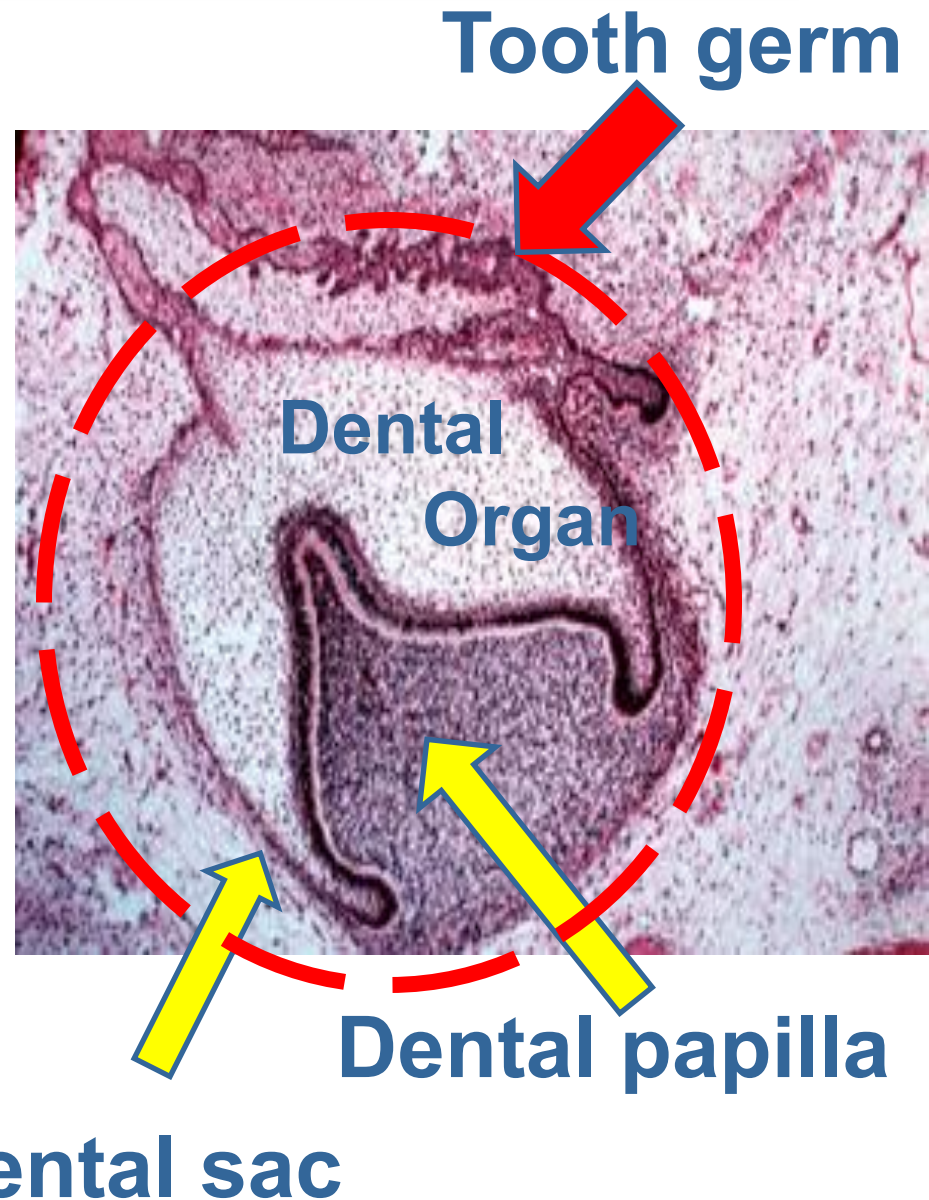


Life History of the tooth



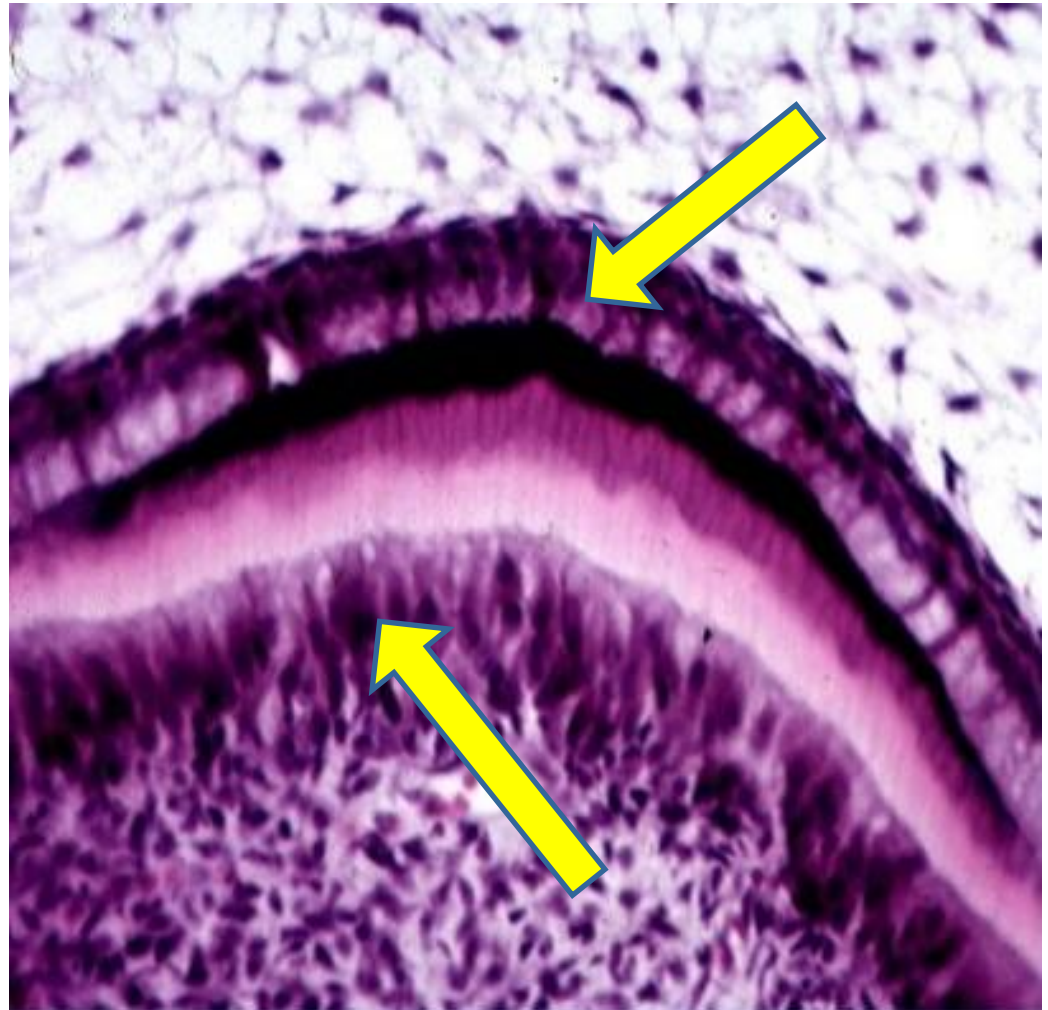
1. Soft growth stage: Dental organ appearance.

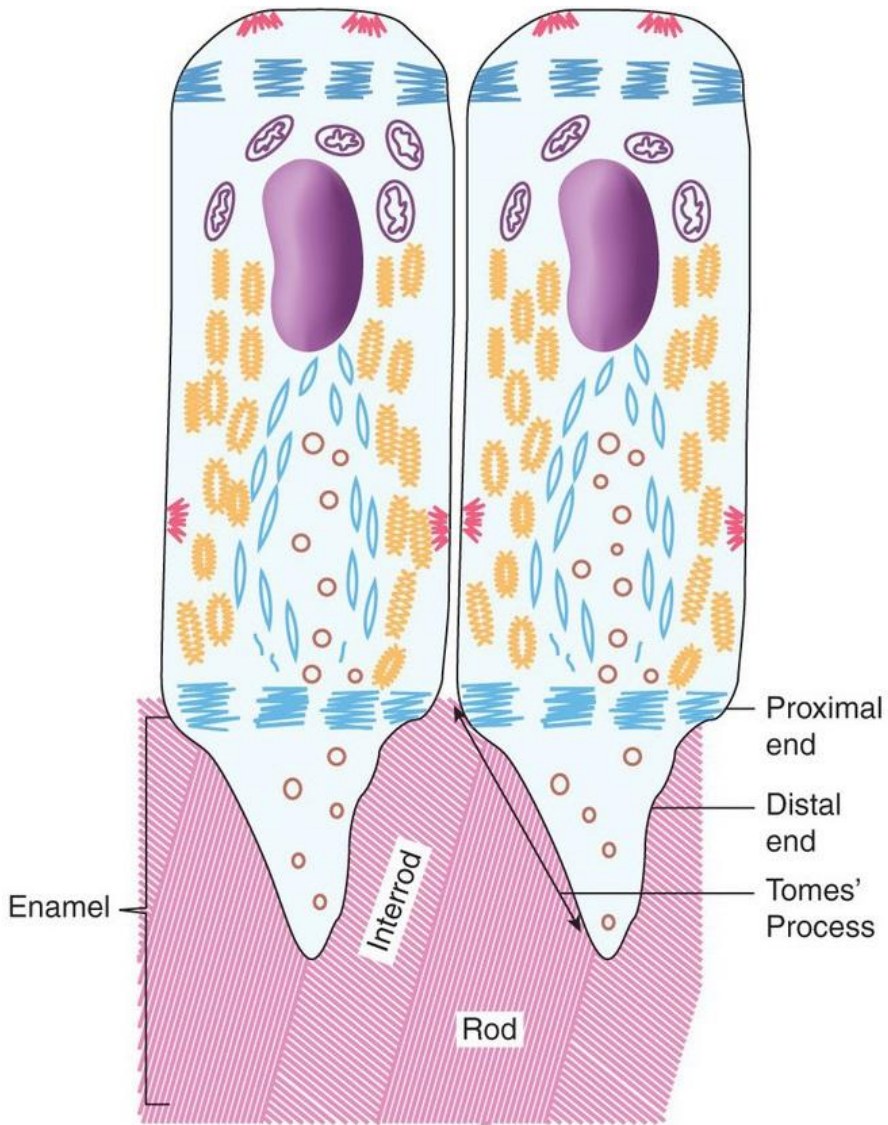
- The **deciduous** & most of **permanent** teeth start their development inside the jaw before birth.
- **Tooth germ** is formed of soft tissue & does not appear in x-ray films.



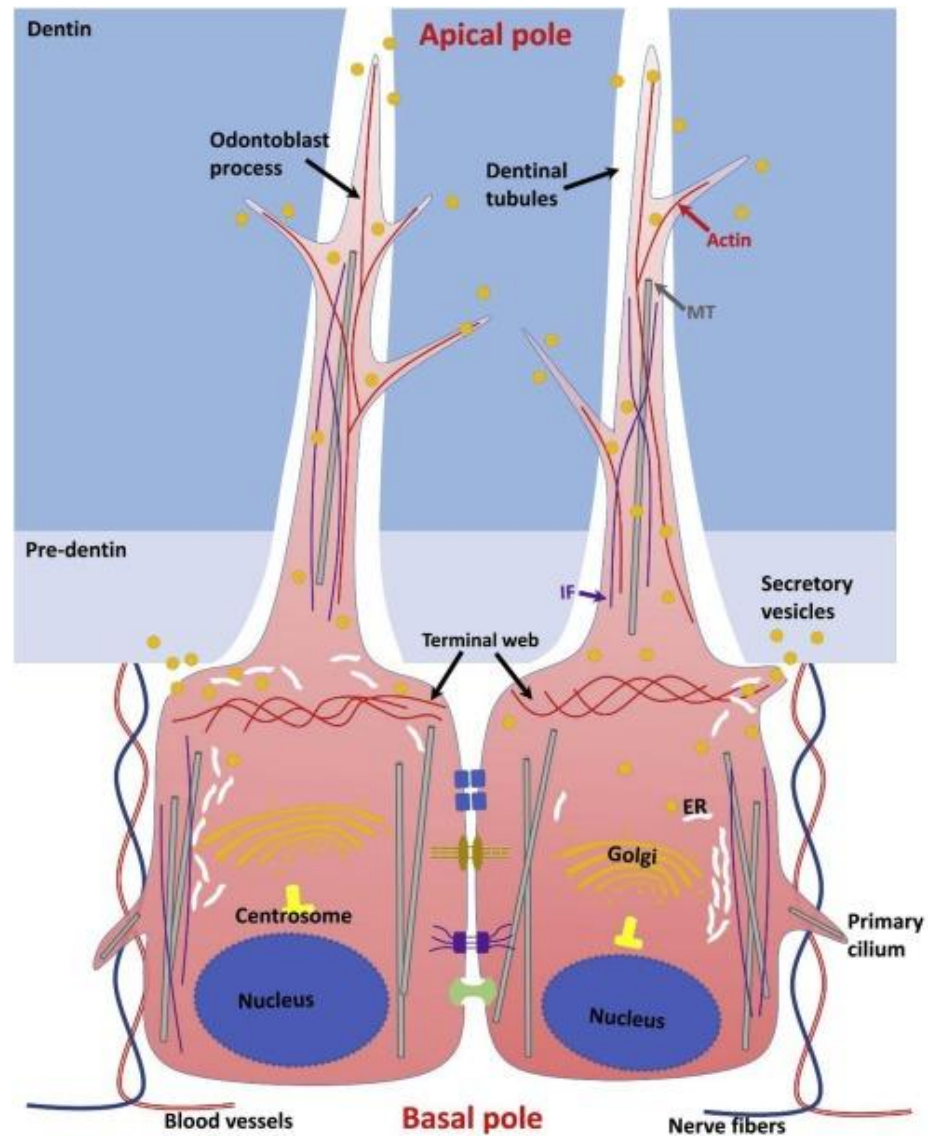
2. Calcification

➤ **Calcification of hard dental tissues (enamel, dentin & cementum) occur by specialized cells.**





Ameloblasts



Odontoblasts

- Mineralized salts are deposited in the soft tissue in the form of small **spherical bodies**, then enlarge to form **lobes** which then fuse to form **calcified** dental tissues.



3. Eruption

- Movement of tooth from crypt to oral cavity.
- Eruption = crown + $\frac{1}{3}$ root.



- **Eruption of mandibular teeth precedes maxillary teeth.**
- **Teeth erupt in girls earlier than boys.**
- **Teeth in both sides of the jaw erupt at the same time.**



4. Attrition

Physiological loss of the hard dental structure due to continuous friction of teeth with each other during mastication.



5. Shedding of deciduous teeth

Physiological loss of deciduous teeth & their replacement by permanent successors.



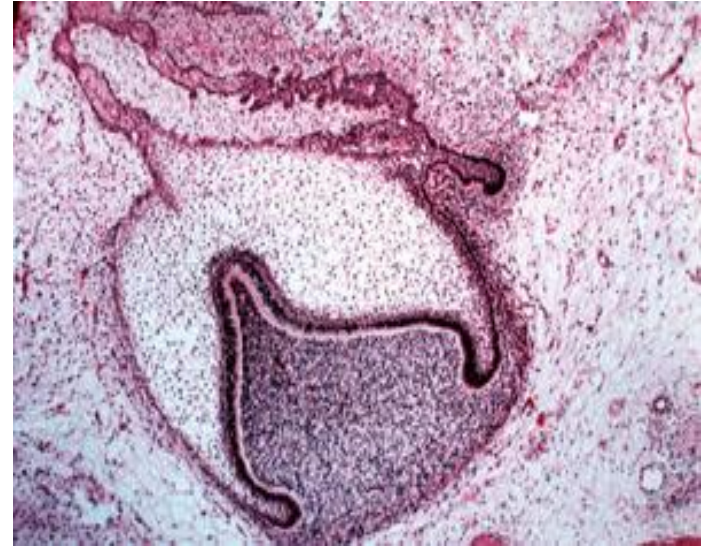
- Before shedding by 3 years, roots begin **resorption** apically till reach their whole length then crowns are exfoliated.



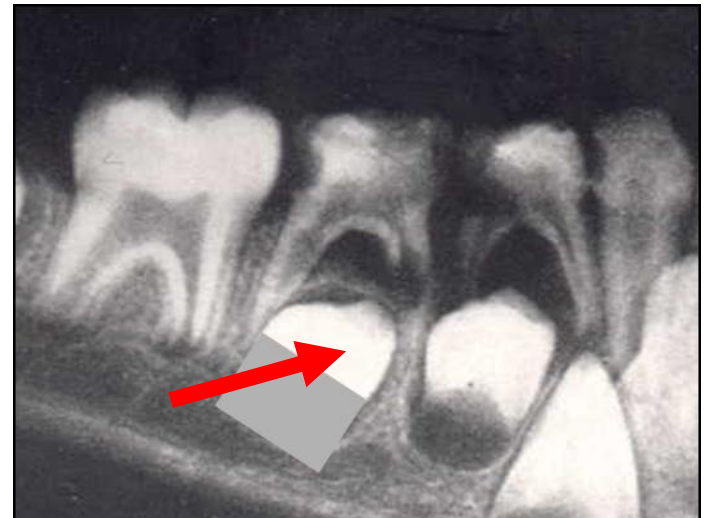
Chronology of a tooth

➤ Order in which a series of events happened.

- Enamel organ appearance.

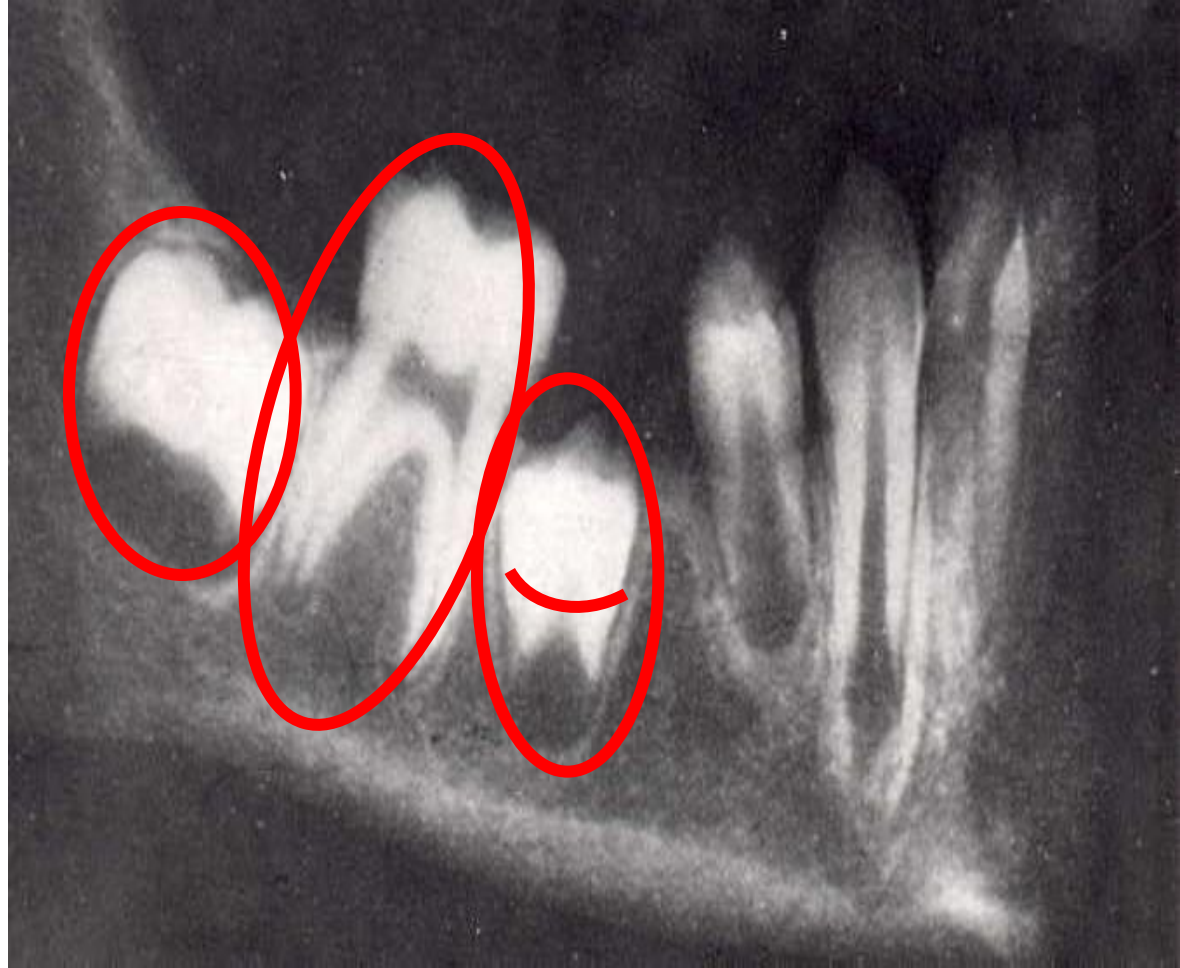


- Beginning of calcification.



- **Crown completed.**

- **Eruption.**



- **Root completed.**

